

Modified Phase Change Microcapsules with Calcium Carbonate and Graphene Oxide Shells for Enhanced Energy Storage and Leakage Prevention

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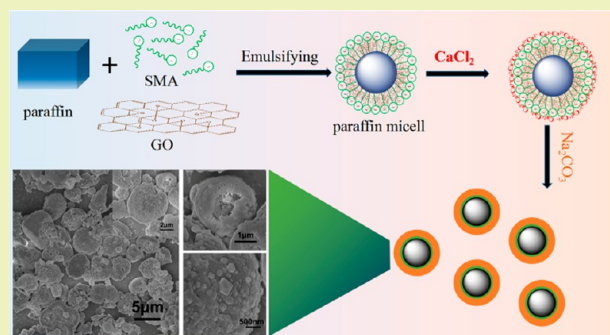
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Supporting Information

ABSTRACT: Environmentally friendly microencapsulated phase change materials (MEPCMs) with calcium carbonate (CaCO_3) shells were modified with graphene oxide (GO), and the effects of GO content and methodology on MEPCMs were examined. The core-shell structure of MEPCMs and crystal structure of CaCO_3 shells were confirmed by scanning electron microscopy (SEM), Fourier-transform infrared spectroscopy (FTIR), and X-ray diffractometer (XRD). The thermal properties and stability of MEPCMs were investigated by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA), suggesting that the addition of GO contributed to improving the heat storage capacity and thermal stability of MEPCMs. When the GO content was 1.0 wt %, the encapsulation ratio of MEPCMs was as high as 73.19%, and the leakage rate was reduced by 89.6% compared to the MEPCMs without GO. Furthermore, the thermal conductivity and mechanical properties of GO modified MEPCMs were improved significantly. The considerable latent heat storage, thermal stability, thermal conductivity, leakage-prevention, and mechanical properties of GO modified paraffin@ CaCO_3 MEPCMs offer potential in green energy applications.

KEYWORDS: Microencapsulated phase change materials, Calcium carbonate, Graphene oxide, Encapsulation ratio, Leakage rate



INTRODUCTION

In recent years renewable energy has become a hot topic because of energy shortages and global warming caused by excessive use of fossil fuels. Phase change materials, as an energy-saving and environmentally friendly energy storage material, can absorb energy from the environment and release again during its phase transition. Its temperature remains unchanged in the whole phase change process, which has attracted a wide range of interests. Paraffin, as a kind of typical organic phase change material, possesses some outstanding merits, such as wide phase change temperature, high heat storage capacity, and no supercooling, which has been widely used in building, clothing, aerospace, military equipment, and so on.^{1–5} However, some disadvantages of the organic solid–liquid phase change materials, such as low thermal conductivity and easy leakage have limited its commercial applications.

Many efforts have been made on the phase change microcapsules with a polymer shell, such as polystyrene (PS),⁶ poly(methyl methacrylate) (PMMA),⁷ melamine-formaldehyde resin (MF),⁸ high density polyethylene (HDPE),⁹ and urea-formaldehyde resin (UF).¹⁰ Although the polymer shells have many advantages, such as high coverage and good compactness,

their poor thermal and chemical stability, low thermal conductivity, and mechanical strength should be further improved. The phase change materials have been reported to be coated with inorganic materials such as silicon dioxide (SiO_2),¹¹ zirconium dioxide (ZrO_2),¹² and calcium carbonate (CaCO_3) for enhancing their thermal conductivity. The MEPCMs with CaCO_3 shells have become a new, environmentally friendly, phase change microcapsule because of low cost, high thermal conductivity, and perfect thermal and chemical stability, etc. Yu et al.¹³ have fabricated MEPCMs with *n*-octadecane cores and CaCO_3 shells, and their encapsulation efficiencies are 21.89–40.04%.¹⁴ Wang et al.¹⁵ have prepared MEPCMs with binary cores and CaCO_3 shells, and their encapsulation efficiencies are 36.6–55.7%. Though the phase change temperature is flexible, the encapsulation ratio is unsatisfactory. Fang et al.¹⁶ have prepared a series of *n*-tetradecane@ CaCO_3 MEPCMs by changing the core/shell mass ratio and mass ratio of the two emulsifiers sodium dodecyl

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paraffin + SMA

GO

Emulsifying

paraffin micelle

CaCl_2

Na_2CO_3

Schematic structure of SMA:

C(c1ccccc1)CC(=O)C(=O)C(c2ccccc2)CC(=O)C(=O)C(c3ccccc3)CC(=O)C(=O)C(c4ccccc4)CC(=O)C(=O)C(c5ccccc5)CC(=O)C(=O)C(c6ccccc6)CC(=O)C(=O)C()[illegible]

Preparation of MEPCMs by Different GO Content. In our experiment, the SMA solution was prepared with 1.25 g SMA, 0.25 g

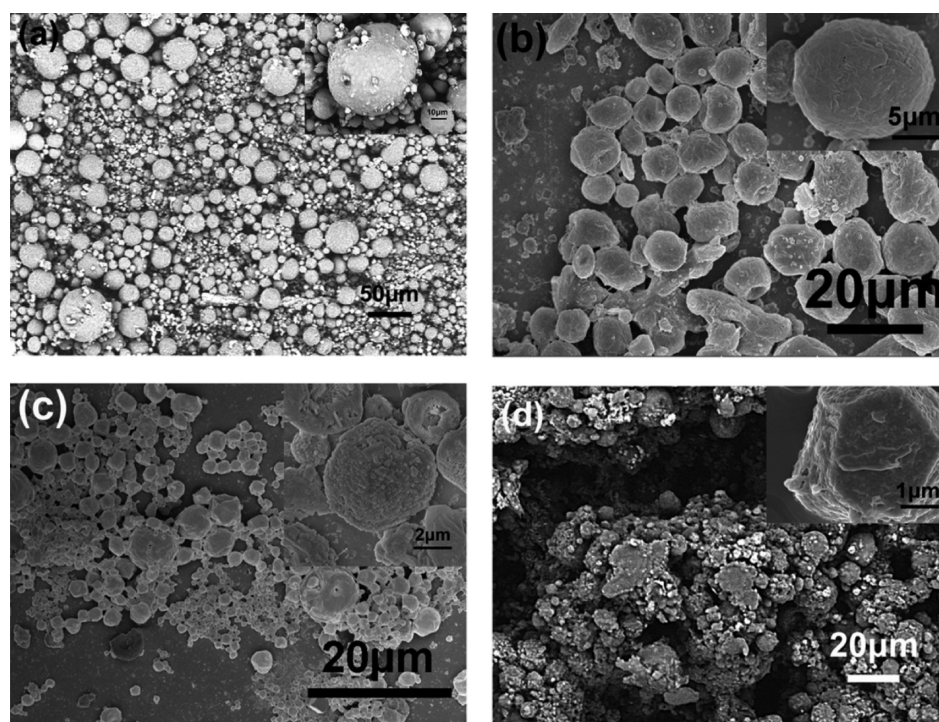


Figure 1. SEM images of (a) P-00, (b) P-05, (c) P-10, and (d) P-15.

sodium hydroxide, and 62.5 mL distilled water, and 5.55 g CaCl_2 was dissolved in 60 mL water to obtain CaCl_2 solution. Na_2CO_3 solution was prepared with 5.3 g Na_2CO_3 and 60 mL water.

The experiment was placed in a constant temperature water bath at 60 °C. SMA solution and 15 mL GO suspension were added into a 150 mL tall beaker, and citric acid solution was added to adjust the pH of mixed emulsifiers to 6. Then 10 g paraffin was added in and emulsified with an emulsifying machine (BRT17-B25) at a rate of 10 000 rpm for 5 min. Then the emulsion was transferred into a 250 mL three-necked flask, and CaCl_2 solution was added dropwise, and the mixture was stirred at a rate of 600 rpm. After 3 h of reaction, the Na_2CO_3 solution was added dropwise. The mixture was slowly stirred at a rate of 300 rpm for 6 h, then the resulting solution was filtered, and the MEPCMs were washed with warm water (45 °C) three times to eliminate the effect of emulsifier on the characterization analysis. Finally, the MEPCMs were dried in a vacuum oven at 40 °C for 24 h. The schematic for preparing MEPCMs is shown in Scheme 1. MEPCMs were labeled as P-00, P-05, P-10, and P-15 when the concentrations of GO suspension were 0.0, 0.5, 1.0, and 1.5 g/L in preparation of emulsion, respectively.

Preparation of MEPCMs with Different Methodologies for Adding GO. As shown in Scheme 2, the methodology is divided into three categories with the other preparation conditions unchanged:

Method a. Paraffin was emulsified by SMA at the rate of 10 000 rpm for 5 min, followed by adding GO suspension (1.0 g/L) and emulsifying at the rate of 10 000 rpm for 1 min. Then CaCl_2 solution and Na_2CO_3 solution were added just as they were for P-10. The obtained MEPCM was noted as P-10a.

Method b. Paraffin was emulsified by SMA at 10 000 rpm for 5 min. After CaCl_2 solution was added and reacted for 3 h at the rate of 600 rpm, GO suspension (1.0 g/L) was added dropwise and reacted for 2 h with a constant stirring rate, then Na_2CO_3 solution was added, and further processing was like method a. The obtained MEPCM was noted as P-10b.

Method c. Paraffin was emulsified with SMA at 10 000 rpm for 5 min. After CaCl_2 solution was added and reacted for 3 h at the rate of 600 rpm, the mixed solution of GO suspension (1.0 g/L) and Na_2CO_3 solution was added dropwise and reacted for 6 h at the rate of 300 rpm. The resulting microcapsules were treated in the way described in method a. The obtained MEPCM was noted as P-10c.

Characterization. The morphologies of the MEPCMs were obtained using TESCAN MAIA3 (Czech) scanning electronic microscope (SEM) and Hitachi TM-3000 (Japan) scanning electronic microscope.

A digital microscope Phenix XSP-35TV-640 was used to take the photographs of emulsion droplets and monitored the formation progress of MEPCMs during synthesis process.

The X-ray diffraction (XRD) patterns of the MEPCMs were using a PANalytical X'Pert PRO X-ray diffractometer (40 kV, 40 mA) equipped with a copper anode (Cu $K\alpha$ radiation, $\lambda = 1.54187 \text{ \AA}$). The measurements were performed using a 2θ scan ranges from 3 to 80°.²⁸

The chemical structures of the paraffin and MEPCMs were analyzed by Fourier transform infrared (FTIR) spectroscopy (Nicolet 5700, USA) with the KBr sampling sheet.

The latent heats of the paraffin and MEPCMs were measured with differential scanning calorimeter (TA Q2000, USA), and all the samples were tested with the rate of 5 °C/min under the protection of a nitrogen atmosphere. Thermogravimetric analysis (TGA) and derivative thermogravimetry (DTG) was evaluated by a thermal analyzer (TA SDT Q600, USA) and measured at the rate of 10 °C/min.

The leakage rate (Lr) was used to express leakage prevention performance of the MEPCMs.²⁹ The test was carried out at 50 °C. Each sample was put on the filter papers, respectively. The sample was weighed by a Mettler Toledo L104 scale every 30 min after being put into the oven, and the weight of the sample was expressed as M_t . The initial mass is M_0 . After each weighing, the filter paper was changed. The leakage rate is calculated as formula 1 below:

$$\text{Lr(\%)} = \frac{M_0 - M_t}{M_0} \times 100\% \quad (1)$$

The thermal conductivity of MEPCMs and paraffin was measured using a thermal conductivity meter (C-Therm TCi, Canada). The tests were performed at 25 °C, and each sample was tested four times and then averaged.

The breakage rate (Br) was used to express the mechanical properties of the MEPCMs.³⁰ About 1 g of microcapsules was weighed to disperse in anhydrous ethanol and centrifuged at 8000 rpm for 2, 6, 10, and 20 min. Then, the samples were washed by deionized water and dried. The

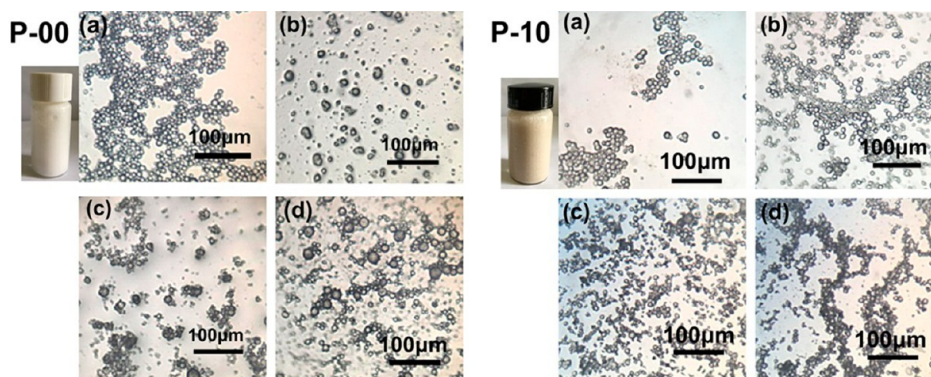


Figure 2. Optical microscope images of P-00 and P-10 in the preparation process. (a) Paraffin emulsion droplets. (b) Microparticles after adding CaCl_2 solution for 3 h. (c) Initial microcapsules after adding Na_2CO_3 solution. (d) Resulting microcapsules after 6 h precipitation reaction.

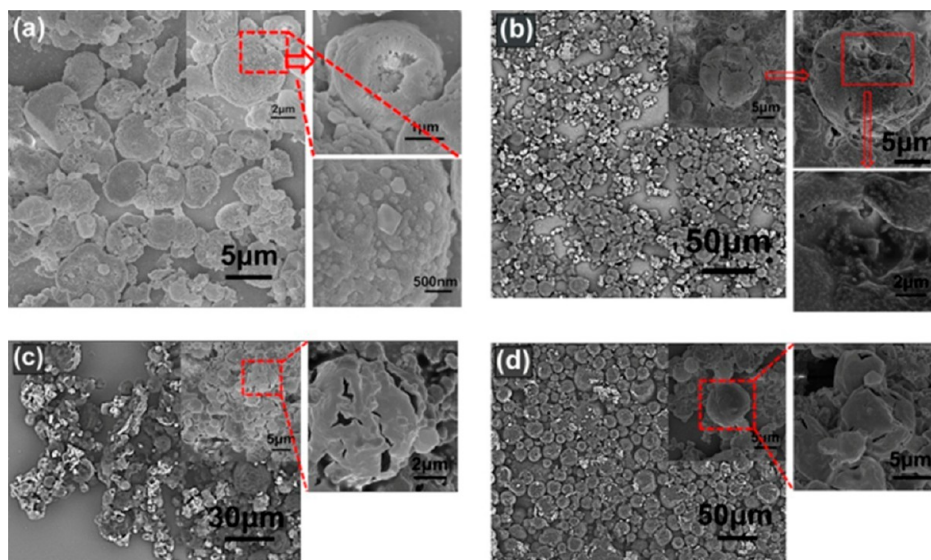


Figure 3. SEM images of (a) P-10, (b) P-10a, (c) P-10b, and (d) P-10c.

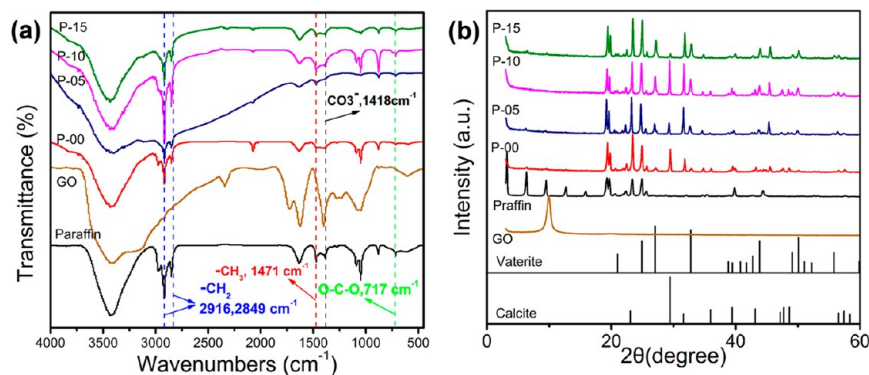


Figure 4. (a) FTIR spectra and (b) XRD patterns of paraffin, GO, P-00, P-05, P-10, P-15, vaterite CaCO_3 , and calcite CaCO_3 .

initial mass and the mass after centrifugation were labeled as M_i and M_n . The breakage rate was calculated as the following eq 2:

$$\text{Br}(\%) = \frac{M_i - M_n}{M_i} \times 100\% \quad (2)$$

RESULTS AND DISCUSSION

Effects of the Preparation Conditions on Morphology and Leakage Prevention Performance of MEPCMs. Content of GO Suspension. In the preparation process of

microcapsules P-00, P-05, P-10, and P-15, paraffin is emulsified with anionic surfactants SMA and GO as coemulsifiers to obtain negatively charged paraffin droplets. Although the GO sheets are negatively charged, GO contains a large number of aromatic rings, which can form a stable solution with the SMA through the π - π stacking interaction. It can be seen from Figure 1 that the surface of P-00 is relatively smooth, and the particle size distribution is not uniform. After adding GO, the surface of microcapsules tend to be more coarse. With the increase of GO content, the roughness of the microcapsule surface increases.

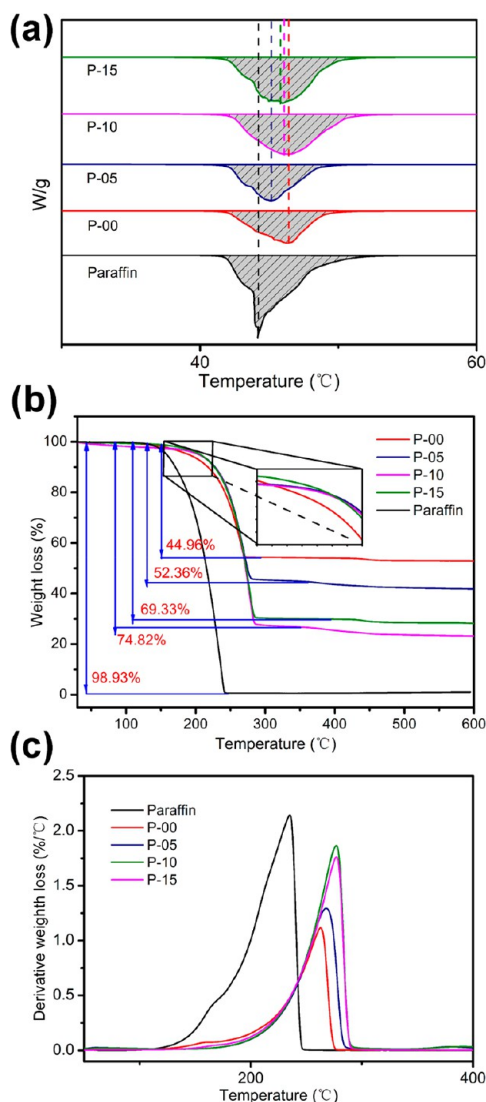


Figure 5. (a) DSC melting curves, (b) TGA curves, and (c) DTG curves of paraffin and MEPCMs.

Table 1. Phase Change Properties of the Paraffin@CaCO₃/GO MEPCMs

samples	CaCl ₂ /paraffin weight ratio (w/w)	melting process		Er (%)
		<i>T_m</i> (°C)	ΔH_m (J/g)	
paraffin		43.85	251.4	
P-00	1:2	42.32	111.6	44.39
P-05	1:2	42.12	129.8	51.66
P-10	1:2	42.29	171.9	68.38
P-15	1:2	43.19	184.0	73.19

When the GO content is 1.0 wt %, CaCO₃ particles can be found at the surface of MEPCMs and most of the microcapsules are covered well without obvious damage. The microcapsule shell still remains in a good spherical state even it is broken for paraffin leakage. The formation process of P-00 and P-10 as shown in Figure 2 can be observed by the optical microscope. The addition of GO has no obvious effect on the size of paraffin droplets. After CaCl₂ solution is added, the particle distribution of P-10 is more uniform. P-00 has better particle dispersity compared to P-10, and its particle size increase evidently. After adding Na₂CO₃

Table 2. Encapsulation Ratio of MEPCMs in This Study and Other Literature

MEPCM	core/shell (mass ratio)	Er (%)	ref
paraffin wax/PS	not mentioned	51.68	6
<i>n</i> -octadecane@PMMA	4:1	89.5	7
<i>n</i> -pentadecane@SiO ₂	3:1	29.8	11
<i>n</i> -eicosane@CaCO ₃	1:1	37.93	13
<i>n</i> -octadecane@CaCO ₃	1:1	40.35	14
RT28+RT42@CaCO ₃	2:1	59.4 ^a	15
<i>n</i> -tetradecane@CaCO ₃	1:1	25.86	16
paraffin@SiO ₂ /GO	about 2:1	50.8 ^b	22
<i>n</i> -dodecanol@MF/GO	about 1:1	about 90	23
<i>n</i> -hexadecane@PS/GO	about 3:1	78	19
paraffin/GO/MF	about 3:1	93.9	24
paraffin@CaCO ₃ /GO	2:1	73.19	this study

^aFormula 4, $Er(\%) = \frac{\Delta H_c}{\Delta H_{c,PCM}} \times 100\%$, *c* is for crystallization process.

^bFormula 5, $Er(\%) = \frac{\Delta H_{m,Composite} + \Delta H_{f,Composite}}{\Delta H_{m,Paraffin} + \Delta H_{f,Paraffin}} \times 100\%$, *m* is for melting latent heat, *f* is for freezing latent heat.

solution, microparticles are gathered around the microcapsules of P-00 and P-10. When the precipitation reaction is completed, the particle size of the microcapsules is significantly increased. Compared to P-00, the particle size of the P-10 is more uniform and tends to be reunited.

The leakage rate of MEPCMs is measured in the oven at 50 °C. It is 14.87%, 3.04%, 1.55%, and 3.57% for the GO content of 0, 0.5, 1.0, and 1.5 wt %, respectively. Considering the effects of GO content on the morphology and leakage rate of MEPCMs, GO with 1.0 wt % contributes to the synthesis of perfect MEPCMs.

Methodology of Adding GO. In the preparation process of P-10a, the paraffin is first emulsified with SMA to obtain the negatively charged paraffin droplets. Then the GO suspension is added into the mixture. The negatively charged GO will repel with the negatively charged paraffin droplets, so there is just a little GO sheets adhered onto the paraffin droplets. In the preparation process of P-10b, GO suspension is added after the paraffin particles adsorb with Ca²⁺ on the surface, and Ca²⁺ and the negatively charged GO are attracted to each other by electrostatic interaction. After adding Na₂CO₃ solution, the CaCO₃ shell is formed. The preparation of P-10c is based on the similar principle of P-10b. As shown in Figure 3, obvious cracks can be seen at the surface of P-10a, P-10b, and P-10c. It could be the methodology in the preparation of P-10a, P-10b, and P-10c cause the cracks. In the process of the preparation of P-10a, P-10b, and P-10c, GO does not act as a coemulsifier. Most of GO sheets are distributed on the surface of microcapsules rather than in the paraffin droplets. In the shell formation process, the GO sheets and CaCO₃ particles can form an interconnected network structure. However, the GO sheets on the surface of the microcapsule are thin, so the thin shells of microcapsules are prone to cracking.

The leakage rate of MEPCMs is measured in the oven at 50 °C. The leakage rate of P-10a, P-10b, and P-10c is 13.55%, 6.83%, and 4.81%, respectively, which is increased sharply in comparison with P-10. Overall, adding GO as a coemulsifier is optimal.

Characterization and Performance of the MEPCMs. *Chemical Composition and Crystallography.* The core-shell structure and chemical composition of MEPCMs are confirmed by FTIR and XRD techniques. As shown in Figure 4a, peaks at 1732, 1623, and 1400 cm⁻¹ correspond to the characteristic peak

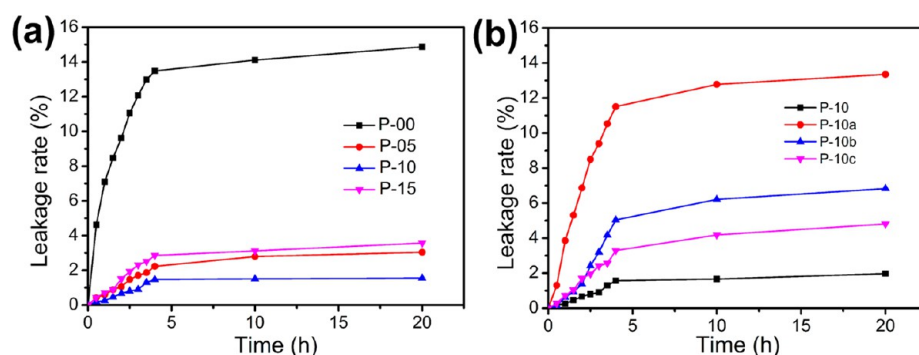


Figure 6. Curve of leakage rate over time. (a) Leakage rate of P-00, P-05, P-10, and P-15. (b) Leakage rate of P-10, P-10a, P-10b, and P-10c.

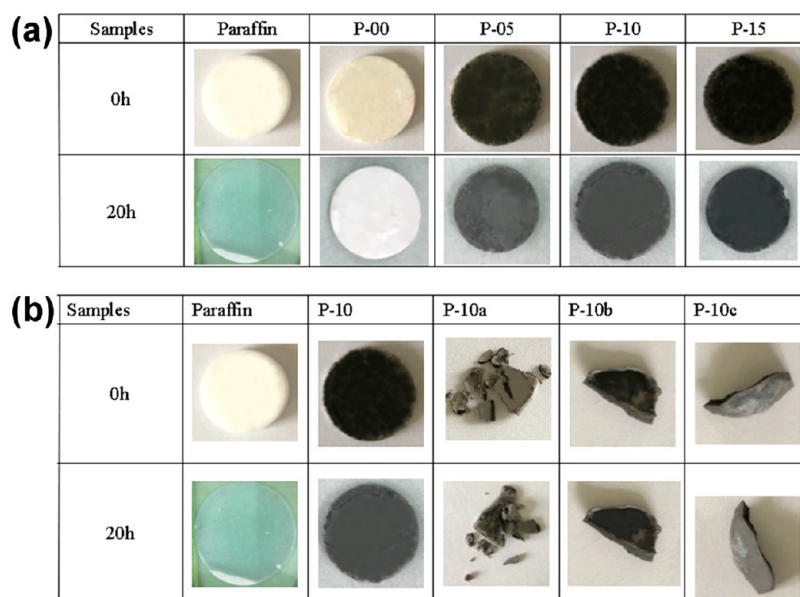


Figure 7. Shape stability of (a) paraffin, P-00, P-05, P-10, and P-15 and (b) paraffin, P-10, P-10a, P-10b, and P-10c deposited at 50 °C.

of GO. And peaks at 1276 and 1047 cm^{-1} belong to the asymmetrical and symmetrical stretching vibration of epoxy bond. The vibration bands at 1471, 2849, and 2916 cm^{-1} confirm the presence of $-\text{CH}_2$. The aborted peak at 879.61 cm^{-1} belongs to the out-of-plane bending vibration of carbonate ions. Furthermore, the characteristic peaks at 717 and 1418 cm^{-1} correspond to the in-plane bending vibration of $\text{O}-\text{C}-\text{O}$ in CaCO_3 and asymmetrical stretching vibration of CO_3^{2-} , respectively. The FTIR spectra of P-10a, P-10b, and P-10c are shown in Figure S1, and the characteristic peak of paraffin and CaCO_3 are marked on the diagram, which suggests that the microcapsules contain both paraffin and CaCO_3 . It can also be seen that the peak of GO is difficult to be found in the curves of MEPCMs, indicating that the low content of GO does not affect the structure of MEPCMs.

Figure 4b shows XRD patterns of MEPCMs, paraffin, GO, vaterite CaCO_3 (JCPDS cards, 01-081-2027) and calcite CaCO_3 (JCPDS cards 00-024-0030). There is only one broad band at $2\theta = 6.91-17.23^\circ$ existing in the XRD pattern of GO. The peak of GO is not obvious in the patterns of MEPCMs, which is attributed to the low GO content in the MEPCMs. The reflections at 2θ of 19.31°, 19.67°, 23.30°, and 24.85° are assigned to the plane (010), (011), (100), and (111) of paraffin, respectively. It is noted that both calcite CaCO_3 and vaterite

CaCO_3 are appeared in the patterns of MEPCMs, and the diffraction peaks of paraffin are also found.

Thermal Property and Thermal Stability of MEPCMs.

Figure 5 shows DSC, TGA, and DTG curves of paraffin and MEPCMs, and the thermal properties are listed in Table 1. When paraffin is microencapsulated, the melting peaks become narrow. It is noticed that the starting weight loss of MEPCMs is more than that of paraffin, which can be explained by the higher thermal conductivity of MEPCMs. It is also observed that the thermal decomposition temperature of paraffin is lower than that of microcapsules, which can be attributed to the protection of CaCO_3 shell. Gasification temperatures of paraffin and P-00 at maximum weight loss is 237.5 and 275.3 °C, respectively. However, gasification temperatures of P-05, P-10, and P-15 at maximum weight loss is 277.1, 281.7, and 281.6 °C, respectively, indicating that the thermal stability of microcapsules is enhanced by the introduction of GO. The weight loss rate of P-00, P-05, P-10, and P-15 before 400 °C obtained from TGA curves is 44.96%, 52.36%, 69.33%, and 74.82%, respectively. The value is similar to the encapsulation ratio (Er) of MEPCMs obtained from DSC curves, as listed in detail in Table 1. Table 2 provides the comparison of the encapsulation ratios of the phase change microcapsule prepared in this work with those in the literature.

Scheme 3. Possible Mechanism of Permeation through the Shells of (a) P-10, (b) P-10a, (c) P-10b, and (d) P-10c

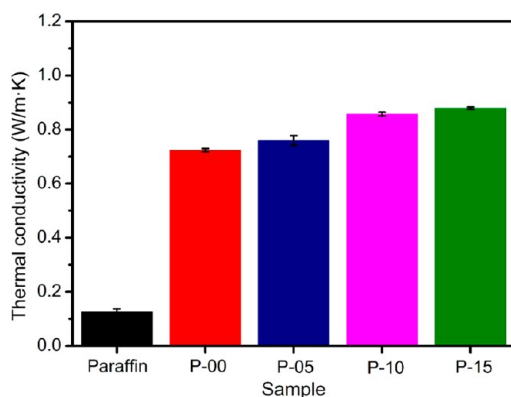
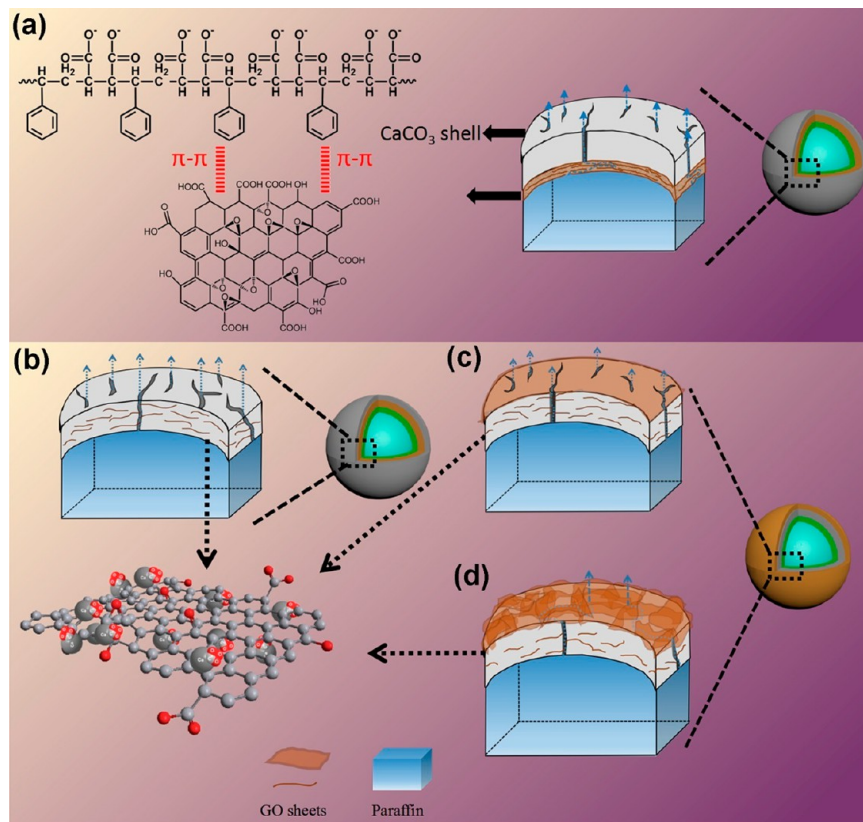


Figure 8. Thermal conductivity of paraffin and MEPCMs with different GO content.

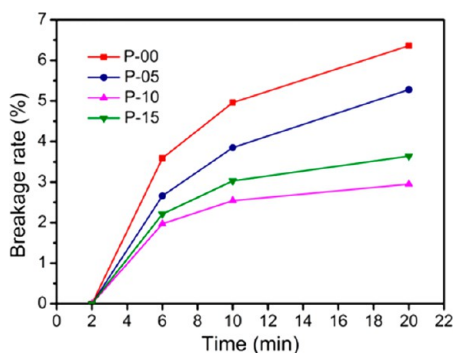


Figure 9. Curve of breakage rate of MEPCMs over time.

$$\text{Er}(\%) = \frac{\Delta H_{\text{m,MEPCM}}}{\Delta H_{\text{m,PCM}}} \times 100\% \quad (3)$$

Leakage Prevention of MEPCMs. Figure 6 shows the leakage rate of MEPCMs at 50 °C for different times. Figure 7 shows the shape of samples before and after heating. The paraffin is completely melted after heating for 40 min and is absorbed by the filter paper. As shown in Figure 6a, the leakage rate of MEPCMs modified with GO is significantly decreased. Compared with MEPCMs without GO, the leakage rate of microcapsules is decreased by 79.6%, 89.6%, and 76.0% when the GO content is 0.5, 1.0, and 1.5 wt %, respectively. This is due to π - π stack interaction between GO and SMA, which forms a protective layer on the surface of paraffin droplets as shown in Scheme 3a. It is found that the leakage rate of P-10a, P-10b, and P-10c is much larger than that of P-10, which is attributed to the different methodology in the preparation of MEPCMs. P-10c has a relatively low leakage rate, because there are more GO sheets on the surface of P-10c. The possible permeation schematic of P-10, P-10a, P-10b, and P-10c is provided, as shown in Scheme 3. The exudation of paraffin needs to pass through long path, and the barrier rate increases. It can be observed that although obvious cracks appear at the surface of P-10a, P-10b, and P-10c, the leakage rate is lower than that of the P-00 without GO. This is due to the fact that the GO layer is adhered into the surface of these microcapsules modified with GO, and in turn, it contributes to obstructing the leakage of paraffin.

Thermal Conductivity of MEPCMs. The thermal conductivity of phase change microcapsules modified with different GO contents is shown in Figure 8. Table S1 shows the specific values of the thermal conductivity. It can be found that the thermal conductivity of the microcapsules is improved

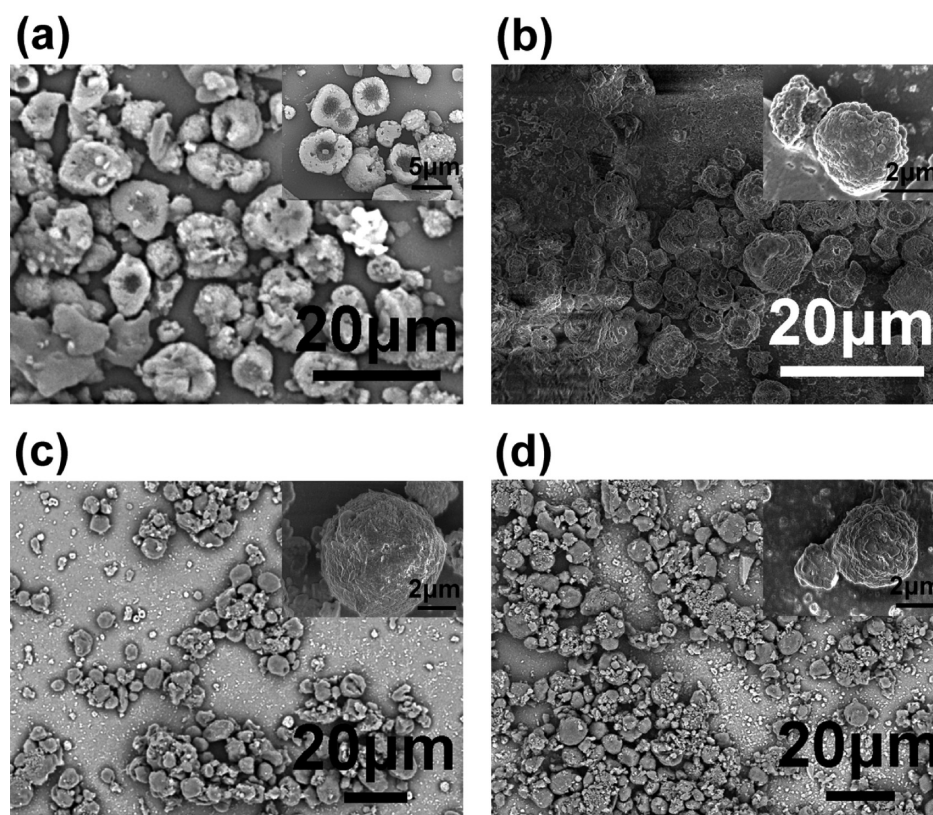


Figure 10. SEM images of (a) P-00, (b) P-05, (c) P-10, and (d) P-15 after breakage test.

remarkably when the paraffin is covered with CaCO_3 as the shell. The high thermal conductivity of the CaCO_3 shell is fully reflected. The thermal conductivity of phase change microcapsules P-00 without GO is $0.724 \text{ W/m}\cdot\text{K}$, which has been greatly enhanced in comparison with the thermal conductivity of phase change microcapsules with polymer shells ($0.08\text{--}0.25 \text{ W/m}\cdot\text{K}$).³¹ With the increase of GO content, the thermal conductivity of microcapsules is increased. The thermal conductivities of microcapsules are 0.759 , 0.857 , and $0.879 \text{ W/m}\cdot\text{K}$ when GO contents are 0.5 , 1.0 , and $1.5 \text{ wt } \%$, respectively. When the GO content is increased to $1.5 \text{ wt } \%$, the increased rate of thermal conductivity is obviously decreased. This is due to the fact that the network structure of GO sheets which can be wrapped and interconnected with the calcium carbonate particles. Then these network structures have a positive effect on the thermal conductivity of the phase change microcapsules. When the GO content becomes large, the network structure of GO sheets tends to be saturated, and a slow increase in rate of thermal conductivity occurs.

Mechanical Properties of MEPCMs. The effect of GO on the damage rate of MEPCMs is shown in Figure 9, and the specific data are listed in Table S2. The breakage rate is 6.36% , 5.28% , 2.95% , and 3.64% corresponding to P-00, P-05, P-10, and P-15, respectively. It is noted that the breakage rate of the MEPCMs with adding GO are decreased in comparison with that without GO, indicating that the mechanical properties of microcapsules modified with GO have been enhanced. When the GO content is $1.0 \text{ wt } \%$, the breakage rate of the MEPCMs is about 54% lower than that of the MEPCMs without GO. This is because GO possesses large surface area and has good dispersion and compatibility with CaCO_3 shells. Additionally, GO can share part of the external force, transfer it, and consume part of the

external energy. Thus, it can improve the mechanical properties of MEPCMs. It is worth noting that when the GO content is increased to $1.5 \text{ wt } \%$, the breakage rate of the MEPCMs is larger than that of $1.0 \text{ wt } \%$, which may result from the agglomeration of GO sheets and CaCO_3 particles. The uneven GO distributions in the CaCO_3 shell results in the high breakage rate. As shown in Figure 10a, after the breakage test, some of the microcapsules have been damaged, the core-shell structure of microcapsules is present, and the core material of the microcapsules has been lost. In Figure 10b, there also exists obvious damage of P-05, but it is significantly improved compared with P-00. Similarly, the core-shell structure of the damaged microcapsules can also be observed, and paraffin has been dissolved in anhydrous ethanol during the breakage test. In Figure 10c and d, however, there are almost no damaged microcapsules; all of the microcapsules are tightly coated with CaCO_3 shells. The results of Figure 10 are consistent with Figure 9 and Table S2.

CONCLUSION

By changing GO content and methodology, a kind of novel phase change microcapsules with CaCO_3 shell modified with GO have been successfully prepared. When GO content is $1.0 \text{ wt } \%$, the MEPCMs with perfect spherical core-shell structures possess a high encapsulation ratio (73.19%), excellent thermal stability and leakage prevention, ideal thermal conductivity ($0.857 \text{ W/m}\cdot\text{K}$), and good mechanical properties. In the preparation process of MEPCMs, when GO is added in the shell, it leads to more cracks on the surface of microcapsules but can still play a positive role in leakage prevention due to its good barrier property. This work presents a new ideas for the modification of phase change microcapsules with GO or other carbon materials, and the

fabricated MEPCMs created here have great potential applications in the areas of energy storage and heat release.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.7b04834.

FTIR spectra data, thermal conductivity data, and breakage rate data (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

MEPCMs, microencapsulated phase change materials; GO, graphene oxide; SEM, scanning electron microscopy; FTIR, Fourier-transform infrared spectroscopy; XRD, X-ray diffractometer; DSC, differential scanning calorimetry; TGA, thermogravimetric analysis; DTG, derivative thermogravimetry; PS, polystyrene; PMMA, poly(methyl methacrylate); MF, melamine-formaldehyde resin; SMA, styrene-maleic anhydride

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